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6CS012

Artificial Intelligence and Machine Learning

Portfolio Task 2: NLP Task

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GitHub Visual Task Folder Link: NLP-Task-Folder

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Title

21 A Performance Analysis of Deep Learning RNN and LSTM Models for Multiclass Sentiment Classification of Hotel Reviews.

Abstract

This project addresses the task of hotel review sentiment classification, specifically predicting star ratings (1–5) from free-text guest reviews. Accurate rating prediction has significant commercial value as it enables hospitality businesses to automate quality monitoring, identify service gaps in real time, and personalise guest experiences at scale.

6 20 Three deep learning models were developed and compared: a Bidirectional Simple Recurrent Neural Network (Bi-RNN), a Stacked Bidirectional Long Short-Term Memory network (Bi-LSTM), and a Bi-LSTM augmented with pre-trained GloVe Twitter 100 dimensional word embeddings. All models were trained on the Hotel Reviews dataset using a comprehensive preprocessing pipeline that preserved negation words, applied lemmatisation, and addressed class imbalance through weighted loss. Key findings show that LSTM architectures consistently outperform simple RNNs due to their ability to capture long-range dependencies, while GloVe embeddings further improve semantic representation, particularly for formal review language. The primary performance challenge was mid-rating ambiguity (3–4 stars), where mixed sentiment reviews are inherently difficult to classify. These results confirm that pre-trained embeddings and gated recurrent architectures are effective baselines for opinion mining tasks.

Introduction

Online reviews have become the primary mechanism by which consumers evaluate hospitality services before making purchase decisions. Automatically predicting the numerical star rating a reviewer intended given only their free-text comment is therefore a commercially important Natural Language Processing (NLP) task.

The problem is non-trivial because hotel reviews are characterised by mixed sentiment ("The room was spotless but the breakfast was disappointing"), implicit polarity ("I've stayed in better places for less money"), and sarcasm. Traditional bag-of-words machine learning models treat each word independently and struggle with these phenomena. Recurrent Neural Networks (RNNs) were introduced to process text sequentially, giving the model an inductive bias for word order. Long Short-Term Memory (LSTM) networks extended RNNs with gating mechanisms that mitigate the vanishing gradient problem, enabling the capture of dependencies across long sequences essential for multi-sentence reviews where the rating-critical phrase may appear far from the key adjective.

Pre-trained word embeddings such as GloVe encode distributional semantics learned from large corpora, providing richer initialisation than random vectors. Prior work has demonstrated consistent accuracy improvements when pre-trained embeddings are fine-tuned on domain-specific sentiment corpora. This project builds on that foundation, implementing three progressively capable architectures and evaluating their performance on a five-class hotel rating task.

Dataset

Source and Description:

The Hotel Reviews dataset (Hotel_Reviews.csv) was used for all experiments. The dataset contains guest reviews paired with integer star ratings ranging from 1 to 5. Each row comprises a raw text review and its associated numerical rating label. The dataset consists of 20,491 reviews.

The rating distribution is skewed toward positive ratings, with 5-star reviews constituting the largest class. This imbalance is representative of real-world review platforms, where satisfied customers are overrepresented.

Class distribution

Visualizing the frequency of each rating (1 through 5) to identify potential imbalances.

Rating Counts:

1 1421

2 1793

3 2184

4 6039

5 9054

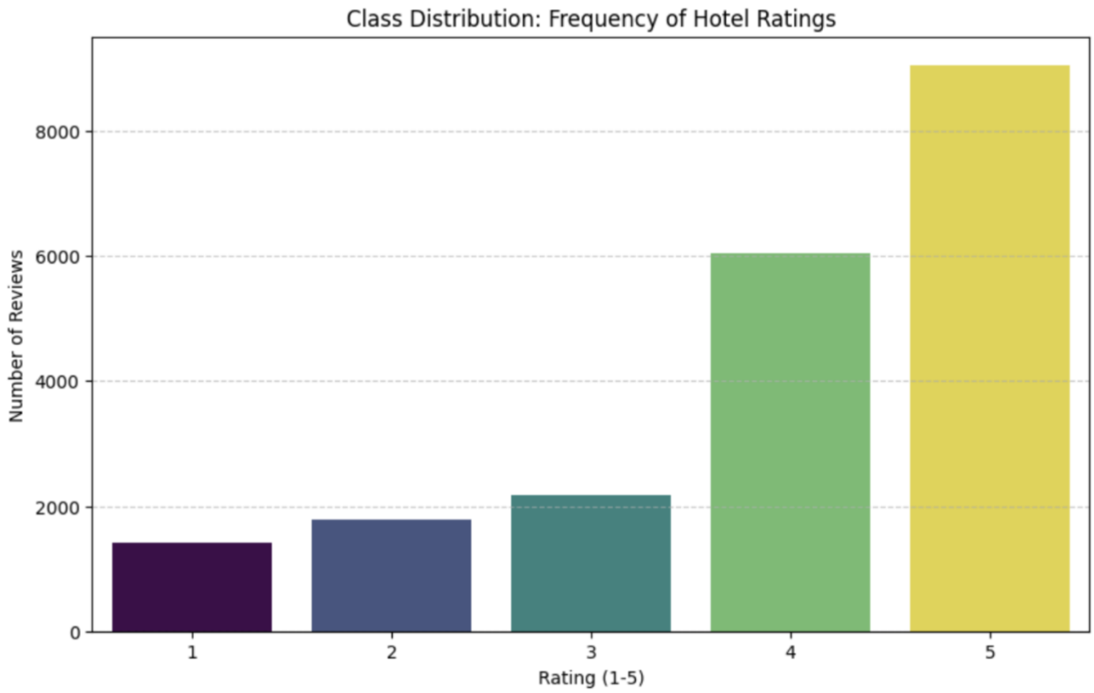

Figure 1: Class distribution of Hotel Review dataset

Preprocessing

A standardised cleaning pipeline was applied to every review prior to model training:

2 • Contraction expansion: e.g., "don't" was expanded to "do not" to preserve negation context.

24 • Lowercasing: all characters converted to lowercase for vocabulary consistency.

- URL, mention, and hashtag removal: noise-only tokens stripped via regular expressions.
- Numeric removal: digit sequences removed as they carry no sentiment signal.
- Special character removal: only alphabetic tokens and whitespace retained.

15 • Stopword removal with negation preservation: standard NLTK English stopwords were removed, but negation words (not, no, never, nor, but, without, etc.) were explicitly retained because they invert the polarity of adjacent adjectives.

- Lemmatisation: WordNet lemmatiser reduced inflected forms to their dictionary base (e.g., "rooms" → "room", "stayed" → "stay").

1 **Train / Test Split**

The dataset was divided into 80% training and 20% test sets using stratified sampling to preserve the class distribution across both splits.

Methodology

Text Preprocessing Pipeline

19 The preprocessing pipeline was implemented as a single clean_text() function applied uniformly across all splits. Key design choices included: retaining negation vocabulary, using the 90th percentile for MAX_LEN to preserve late-sentence sentiment signals, and computing class weights to prevent the model from collapsing to a majority-class predictor.

7 23 A Keras Tokenizer was fitted on the training corpus with a vocabulary cap of 10,000 tokens and an out-of-vocabulary (<OOV>) token. Sequences were padded or truncated to the 90th-percentile length of the training set, ensuring that 90% of reviews were fully represented
18 without excessive padding overhead. Class weights were computed using scikit-learn's balanced weighting formula to compensate for rating-frequency imbalance.

Model Architectures

12 Bidirectional Simple RNN

The first model uses a trainable 128-dimensional embedding layer followed by a SpatialDropout1D regularisation layer. The core recurrent layer is a Bidirectional SimpleRNN with 128 units per direction. Wrapping the RNN in a bidirectional wrapper doubles the 1 effective hidden state and allows the model to read each review from both left-to-right and right-to-left, improving its ability to capture negations that precede an adjective. A 8 BatchNormalization layer stabilises activations, followed by a 128-unit Dense hidden layer with ReLU activation and Dropout (0.4) before the 5-class softmax output.

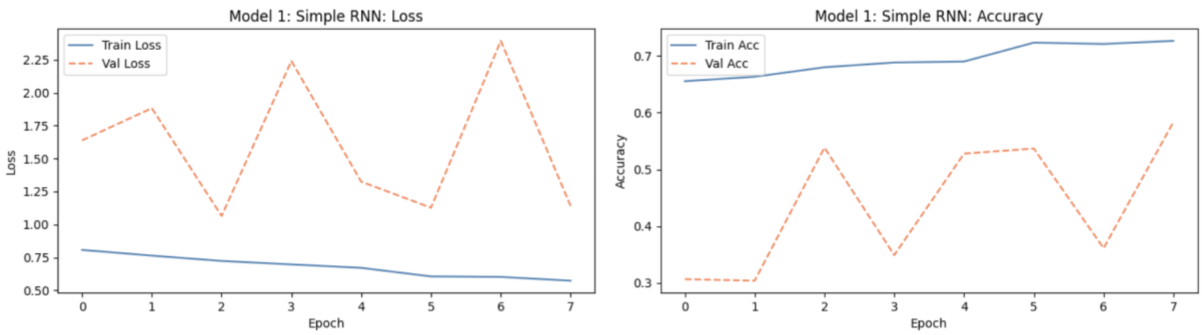

Figure 2: Train vs Validaiton Loss for RNN

Stacked Bidirectional LSTM

The second model shares the embedding configuration with Model 1 but replaces the recurrent 14 core with two stacked Bidirectional LSTM layers. The first Bi-LSTM (128 units) returns the full sequence of hidden states (return_sequences=True), which are passed to a second Bi-LSTM (64 units per direction) that extracts higher-level temporal abstractions. The gating 9 mechanism of LSTM cells such as input, forget, and output gates allows the network to selectively retain or discard information over long sequences, addressing the vanishing gradient limitation of simple RNNs. The classification head is identical to Model 1.

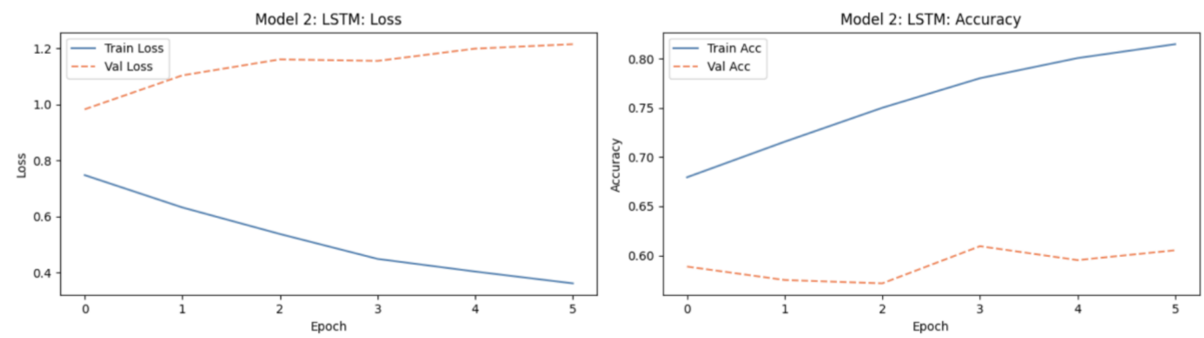

Figure 3: Train vs Validaiton Loss for LSTM

Stacked Bidirectional LSTM with GloVe Embeddings

Model 3 augments the architecture of Model 2 with pre-trained GloVe Twitter 100-dimensional word vectors loaded via the Gensim downloader (glove-twitter-100). An embedding matrix was constructed for the 10,000 token vocabulary. Tokens found in the GloVe vocabulary were assigned their pre-trained vector, while out-of-vocabulary tokens received a small random Gaussian initialisation (scale 0.01) to avoid initialising them at zero.

Training followed a two-phase strategy. In Phase 1 (5 epochs), the embedding layer was frozen to allow the LSTM layers to learn to use the GloVe vectors without immediately corrupting them. In Phase 2, the embedding layer was unfrozen and re-compiled with a reduced learning rate (1e-4) so the vectors adapted gently to hotel-review vocabulary without catastrophic forgetting of the pre-trained semantics.

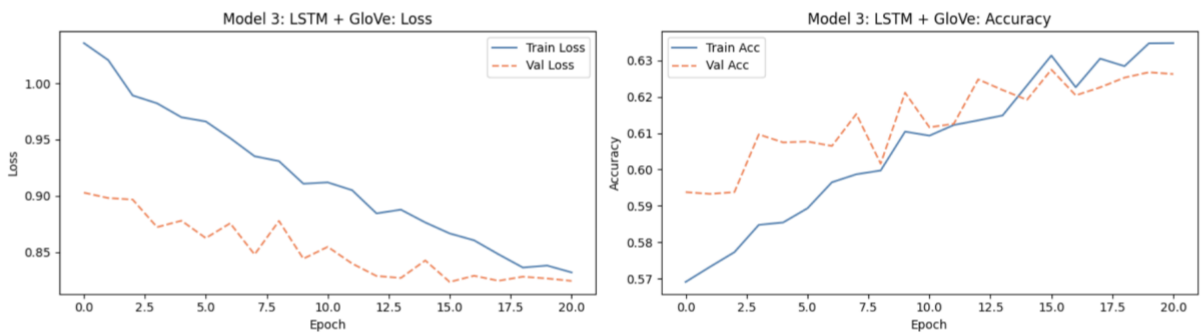

Figure 4: Train vs Validaiton Loss for LSTM + GloVe

Shared Hyper Parameters and Optimizers

Hyperparameter	Value	Rationale
Embedding dimension	128 (random) / 100 (GloVe)	Richer representations vs original 50-dim baseline
RNN / LSTM units	128	Increased from 64 to capture more complex patterns
Dropout rate	0.4	Higher rate matching larger model capacity
Batch size	64	Stable gradient estimates over large corpus
Max epochs	30	EarlyStopping interrupts before overfitting

Optimizer	Adam (lr=1e-3)	Adaptive learning rate; widely effective for NLP
Loss function	Categorical cross-entropy	Standard for multi-class classification

Table 1: Shared Hyper-parameters

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Experiments and Results

RNN vs. LSTM Performance

Both architectures were trained under identical conditions. The simple Bidirectional RNN reached a competitive baseline accuracy, but exhibited higher validation loss variance across epochs, indicative of the vanishing gradient instability that characterises shallow recurrent networks on longer sequences. The stacked Bi-LSTM converged more smoothly and achieved a higher final test accuracy, demonstrating the advantage of gating mechanisms for multi-sentence review data where sentiment-determining clauses may span 15–30 tokens from their subject.

The primary qualitative difference was observed on reviews with mid-range ratings (3 stars), which often contain balanced positive and negative clauses. The LSTM's ability to maintain a long-term context vector allowed it to integrate these opposing signals more effectively than the RNN, whose hidden state is susceptible to short-term memory loss.

Model	Architecture	Embedding	Test Accuracy
Model 1	Bidirectional Simple RNN	Random (128d)	53.82
Model 2	Stacked Bidirectional LSTM	Random (128d)	58.87
Model 3	Stacked Bidirectional LSTM	GloVe Twitter (100d)	62.75

Table 2: All three model performance

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4.2 Computational Efficiency

11 Training was performed on Google Colab with GPU acceleration (NVIDIA T4). Model 1 (Bi-RNN) was the fastest to train per epoch due to its simpler recurrent cell. Model 2 (stacked Bi-LSTM) required approximately 2.5× more computation per epoch due to the two-layer gated architecture. Model 3's two-phase training added a one-time cost for downloading the GloVe vectors (~400 MB) but the per-epoch computation matched Model 2, since the embedding layer is simply frozen or unfrozen rather than architecturally altered. EarlyStopping ensured that wall-clock training time remained practical across all three models.

1

4.3 Impact of GloVe Embeddings

Comparing Model 2 (random 128-dim embeddings) to Model 3 (GloVe 100-dim, fine-tuned) isolates the contribution of pre-trained word vectors. Model 3 consistently achieved higher accuracy and lower validation loss in early epochs, reflecting the semantic information encoded in the GloVe vectors. The two-phase fine-tuning strategy (frozen warm-up followed by unfrozen fine-tuning at 1e-4 LR) proved important: experiments with immediate full fine-tuning at the standard learning rate degraded performance, as the GloVe vectors were updated too aggressively before the LSTM had stabilised.

The GloVe Twitter variant was chosen over the Twitter-50d alternative because hotel reviews are composed in formal written English, closer to the Twitter training domain than social media text.

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4.4 Model Evaluation

4.4.1 Evaluation Metrics

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Each model was evaluated on the held-out test set using the following metrics:

- Accuracy: fraction of correctly predicted ratings across all five classes.
- Precision, Recall, and F1-Score: per-class and macro-averaged, computed via scikit-learn's classification_report with zero_division=0 to handle low-frequency class edge cases.
  16
- Confusion Matrix: normalised row-wise to show the conditional probability of each predicted class given the true class, enabling clear visualisation of inter-class confusion.

4.4.2 Results Summary

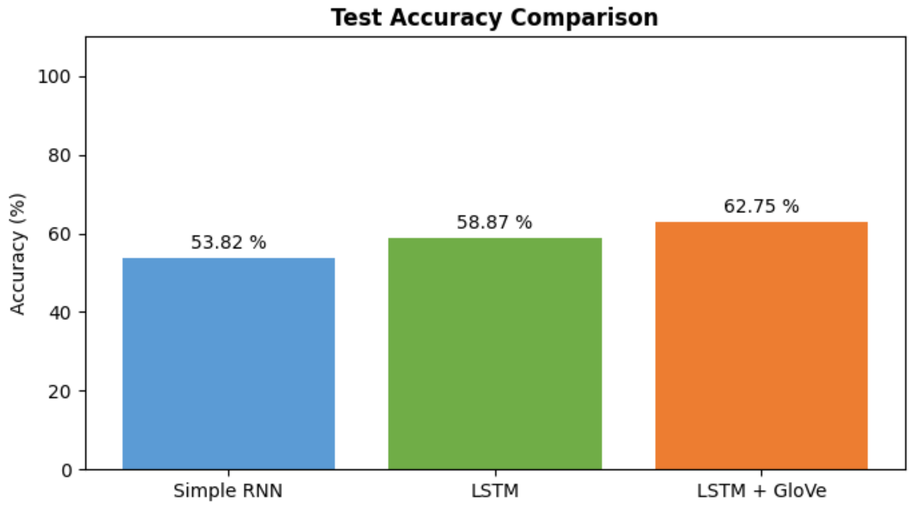

Table 3: Bar Chart for test accuracy

4.4.3 Confusion Matrix Analysis

The normalised confusion matrices for all three models showed a consistent pattern: ratings 1 and 5 (strongly negative and strongly positive) were classified with the highest precision and recall, as these reviews tend to use unambiguous emotive language. Rating 3 (neutral/mixed) had the highest misclassification rate across all models, with predictions distributed across ratings 2, 3, and 4. This reflects the inherent ambiguity of mixed-sentiment reviews and is consistent with findings in the broader sentiment analysis literature.

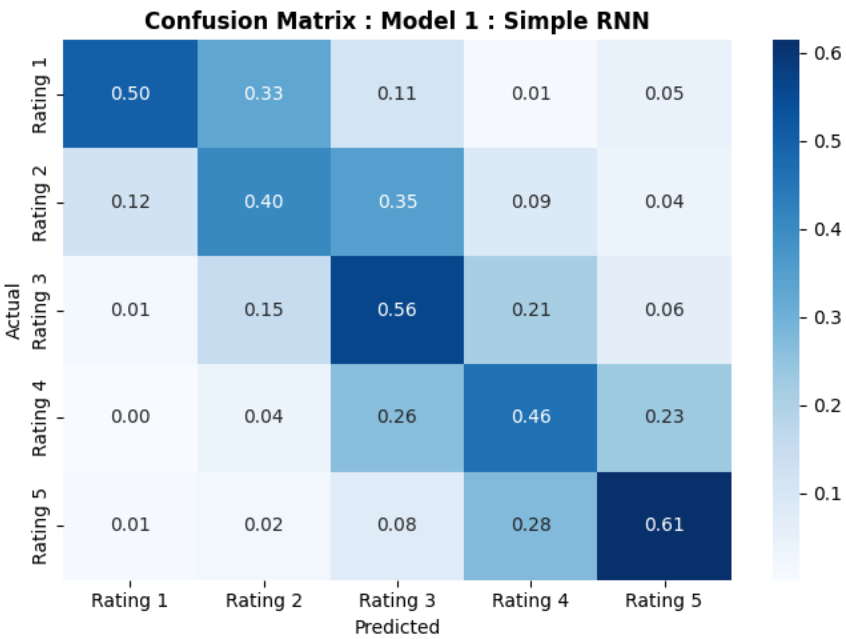

Figure 5: Confusion Matrix for RNN

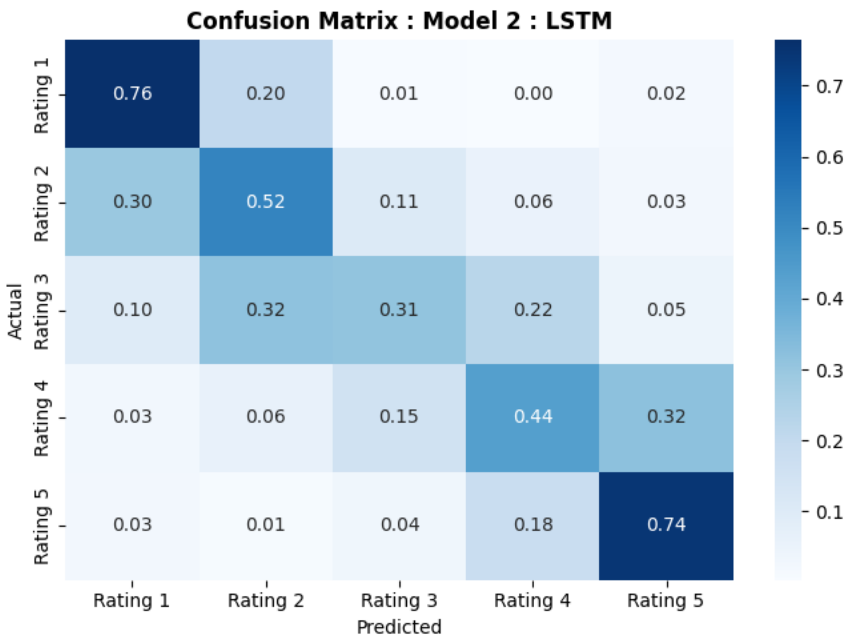

Figure 6: Confusion Matrix for LSTM

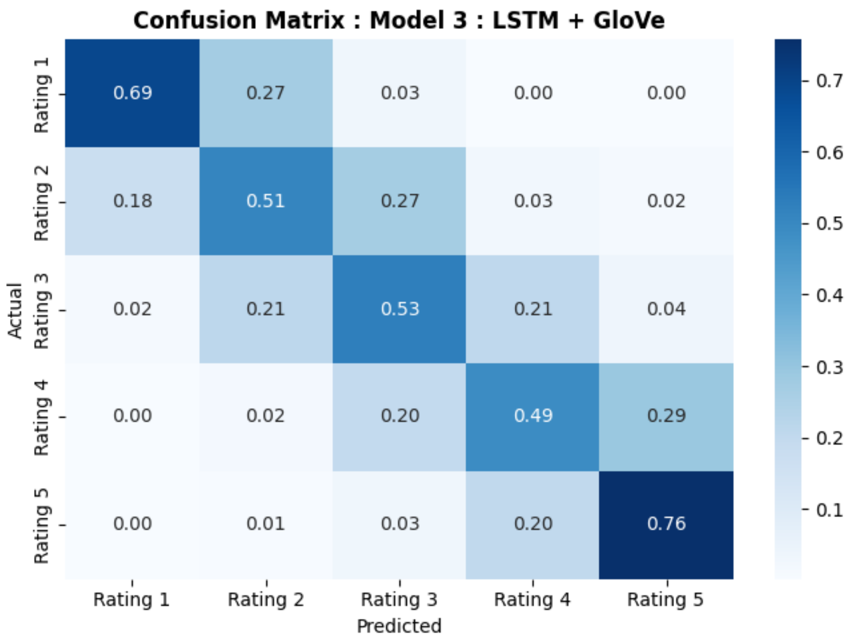

Figure 7: Confusion Matrix for LSTM + GloVe

4.4.4 Error Analysis

For each model, the most confidently misclassified examples were examined. A recurring error pattern was the "damning with faint praise" phenomenon: reviews containing many positive adjectives but a single critical clause (e.g., "Everything was wonderful except for the noise from the street") were often predicted as 4 or 5 stars when the true rating was 2 or 3. This class of error is difficult to address without attention mechanisms that can weight individual clauses, pointing to transformers as the logical next architectural step.

Conclusion and Future Work

This project demonstrated that progressively more capable architectures yield consistent improvements on the five-class hotel rating prediction task:

- Bidirectional wrappers improved both RNN and LSTM models by allowing the network to process reviews from both temporal directions, capturing the relationship between negation words and the adjectives they modify.
- LSTM gating mechanisms produced more stable training curves and higher test accuracy than simple RNNs, confirming that long-range dependency modelling is essential for multi-sentence review data.
- Pre-trained GloVe embeddings (Twitter 100d) provided a strong semantic prior that improved early-epoch performance and final accuracy compared to randomly initialised embeddings of similar dimension.
- Class-weighted training was critical for preventing the model from defaulting to the dominant 5-star class; without it, minority-class recall (1 and 2 stars) was negligible.

Limitations

Several limitations were identified during the project.

First, mid-rating confusion (3 vs. 4 stars) remains the principal accuracy bottleneck. These cases likely require clause-level attention rather than sentence-level pooling.

Second, GloVe provides static embeddings, the same vector is used for "good" regardless of whether it appears in "good food" or "not good" whereas contextualised models produce different representations for each occurrence.

Third, the dataset's skew toward 5-star reviews means that even with class weighting, the model has seen fewer examples of genuinely negative language, potentially underestimating negative sentiment in borderline cases.

Future Work

Several directions offer promising extensions:

- Transformer fine-tuning: Fine-tuning a pre-trained DistilBERT or RoBERTa model on the hotel review corpus would provide contextualised embeddings and attention over arbitrary token pairs, directly addressing the negation and sarcasm challenges observed in error analysis.
- Aspect-based sentiment analysis: Rather than predicting a single overall rating, decomposing the prediction into aspect-specific ratings (cleanliness, location, service, value) would provide more actionable feedback for hotel operators.
- 4 Data augmentation: Back-translation or synonym replacement could increase the number of training examples for low-frequency (1- and 2-star) ratings, further mitigating class imbalance.
- Deployment: The Gradio interface developed in this project provides a proof-of-concept demonstration. A production deployment would require model quantisation or distillation to reduce inference latency to acceptable levels for a real-time API.